NATURAL LANGUAGE PROCESSING • APPLIED ASSET ENGINEERING

Week 6 — Asset Engineering: Reusable NLP Asset

Modular Multitask Pipeline for Document Intent Classification, Role Routing, Information Extraction, and Persona-Conditioned Summarization

Project Repository	`uni-comms-intelligence`
Milestone / Deliverable	**Week 6 — Asset Engineering (Reusable NLP Asset)**
Submission Target Date	16 October 2026
Project Group	**Group 11**
Implemented Interface	**Python Pipeline Module (`IntegratedNoticePipeline`) & CLI (`scripts/predict.py`)**

Executive Summary

In Week 6 (Asset Engineering), the University Communications Intelligence project transitions from exploratory notebooks into a structured, reusable software asset. The asset encapsulates a three-stage NLP pipeline for processing institutional university emails and notices: (1) 5-class document category classification and multi-label audience routing, (2) hybrid information extraction via regex and transformer NER, and (3) audience-gated abstractive summarization using fine-tuned BART-base with entity masking. The asset is exposed as an importable, type-annotated Python module and command-line execution interface, verified on 114 real institutional notices under a clean 80/17/17 split.

Submission Deliverables: `Week_6_Asset_Engineering_Report.docx` · `Week_6_Asset_Engineering_Report.pdf` · `README.md`
Repository: `uni-comms-intelligence` · Evaluation Split: 114 Real Institutional Notices (80 Train / 17 Val / 17 Test)

1. Project Overview & Deliverables

University Communications Intelligence addresses institutional noise by converting unstructured campus circulars and email broadcasts into structured, actionable intelligence. The Week 6 deliverable packages this capability as a reusable Python software asset that: (a) classifies circulars across five operational domains (`academic`, `admissions`, `announcements`, `examinations`, `placements`); (b) routes notices to multi-label recipient roles (`student`, `faculty`, `admin`); (c) extracts structured entities (dates, deadlines, times, contact emails, portal URLs, administrative roles); and (d) synthesizes persona-conditioned summaries for target audiences while applying null fast-paths to untargeted roles.

2. System Architecture & Processing Workflow

The asset executes as a coordinated sequential pipeline with audience-gated control flow (Figure 1). Text enters via raw string, header-parsed email, or extracted PDF text. Following title extraction and normalization, Stage 1 produces category and audience predictions. Crucially, the predicted audience set acts as a runtime gating controller for Stage 3: if an audience persona is not targeted, summarization immediately emits a deterministic null response (*"No specific [role] action stated."*), bypassing generative computation.

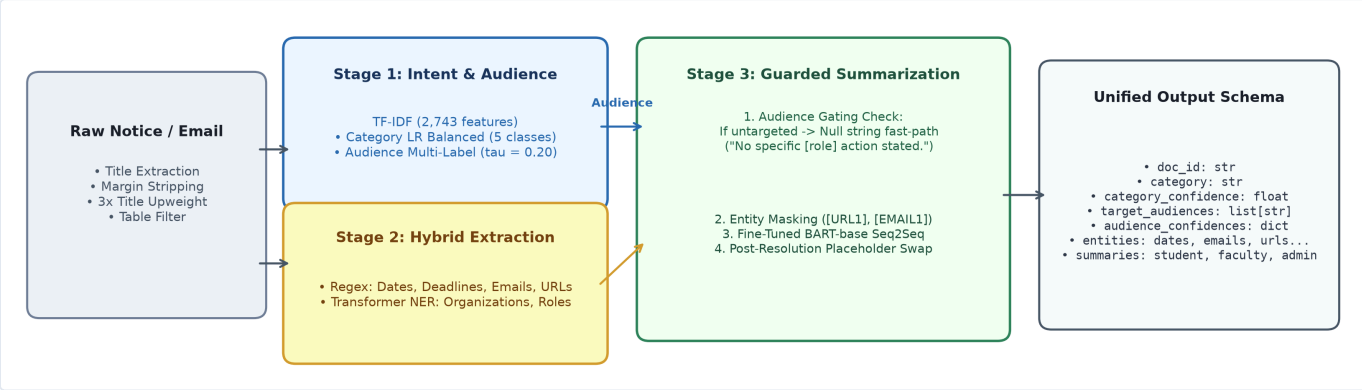


Figure 1: 3-Stage Pipeline Architecture, Audience Gating, and Unified Output Contract

3. Technical Implementation

• **Stage 1 (Category & Audience Routing):** Tokenizes text into 2,743 TF-IDF features (unigrams and bigrams, sublinear TF, `min_df=2`, `max_df=0.9`). Subject lines are upweighted 3x to sharpen jurisdictional cues. Primary category is predicted using class-weighted Logistic Regression (`class_weight='balanced'`) to counteract heavy distribution skew toward examinations and placements. Multi-label audience routing is performed via OneVsRest Logistic Regression with per-role decision thresholds tuned on validation data ($\tau = 0.20$ across all three roles).
• **Stage 2 (Hybrid Information Extraction):** Employs deterministic regexes for unambiguous operational entities: dates/deadlines (`DD/MM/YYYY`, `DD-MM-YYYY`), contact emails (`EMAIL_RE`), external URLs and forms (`URL_RE`), operational times, and canonical administrative designations (`ROLE_CANONICAL`). Transformer NER (`dslim/bert-base-NER`) extracts organizations and person names. Dense numeric seating tables (digit ratio > 0.70) are filtered to prevent memory bloat.
• **Stage 3 (Guarded Persona Summarization):** Uses fine-tuned `facebook/bart-base` Seq2Seq model conditioned on structured prompts: `summarize for {role} | category: {category} | dates: {dates} | notice: {masked_notice}`. To mitigate abstractive URL and email hallucination, real URLs and emails are tokenized to `[URL1]` and `[EMAIL1]` before generation, then swapped back to exact Stage 2 strings post-generation.

4. Interface & Input/Output Specifications

The asset is implemented as an importable Python module (`submission/week_5/integrated_pipeline.py`) and CLI (`scripts/predict.py`). *Note: No active HTTP REST server daemon exists currently (fastapi is declared in pyproject.toml for future deployment).* The primary entry point is the `IntegratedNoticePipeline` class:

```
pipeline.process_document(doc_id: str, raw_text: str, title: Optional[str] = None) -> dict[str, Any]
```

Field Name	Type	Status	Description & Constraints
`doc_id`	`str`	**Input (Req)**	Unique institutional string identifying the document.
`raw_text`	`str`	**Input (Req)**	Full raw notice body, including headers if available.
`category`	`str`	**Output**	Predicted category: academic, admissions, announcements, examinations, placements.
`category_confidence`	`float`	**Output**	Softmax probability score of the top predicted category in [0.0, 1.0].
`target_audiences`	`list[str]`	**Output**	Predicted recipient roles: subset of ['student', 'faculty', 'admin'].
`entities`	`dict`	**Output**	Extracted dates, times, emails, urls, organizations, persons, and roles.
`summaries`	`dict[str, str]`	**Output**	Persona-specific summaries; untargeted roles receive deterministic null strings.

Concise Example Payload (acad-001 output):


```
{
  "doc_id": "acad-001", "title": "ID card wearing is mandatory",
  "category": "announcements", "category_confidence": 0.7000,
  "target_audiences": ["student"], "audience_confidences": {"admin": 0.1820, "faculty": 0.1945, "student": 0.
9912},
  "entities": {"dates": [], "emails": ["academic@nmims.edu"], "urls": ["https://forms.google.com/xyz"], "role
s": ["Academic Coordinator"]},
  "summaries": {
    "student": "Wearing ID card on campus is mandatory per SRB. Biometric entry strictly enforced. Submit que
ries via https://forms.google.com/xyz.",
    "faculty": "No specific faculty action stated.", "admin": "No specific admin action stated."
  }
}
```


5. Execution & Reproducibility Guide

The asset requires Python 3.11+. Dependencies are defined in `pyproject.toml`. All training, evaluation, and inference workflows are executed directly from terminal:

```
# 1. Environment Setup & Dependency Installation
python3.11 -m venv .venv && source .venv/bin/activate && pip install -e .

# 2. Reproduce Dataset Build, Classifier Training & Quantitative Evaluation
.venv/bin/python scripts/build_week6_dataset.py       # Parse 114 emails, isolate leakage pairs, 80/17/17 spl
it
.venv/bin/python scripts/train_week6_classifiers.py   # Retrain 5-class Category and multi-label Audience mod
els
.venv/bin/python scripts/evaluate_week6_classifiers.py # Run held-out test evaluation suite & generate metric
s

# 3. Run Inference via Command-Line Interface (CLI)
.venv/bin/python scripts/predict.py --text "Odd Sem Timetable Re-exam link active until 14th October."
.venv/bin/python scripts/predict.py --file submission/week_6/data/dump/examinations_mails.txt
.venv/bin/python scripts/predict.py --pdf data/raw/pdfs/10.pdf
```

6. Quantitative Evaluation & Benchmarks

All quantitative evaluation figures were measured strictly on the held-out test split of 17 real college emails from SVKM NMIMS MPSTME under a verified leakage-free split (0 synthetic notices, 0 duplicated records).

Table 1: Category Classification Evaluation (5 Classes, Test N = 17):

Metric	Standard Logistic Regression	Class-Weighted Logistic Regression	Impact / Delta
Accuracy	70.59% (12 / 17)	**82.35% (14 / 17)**	**+11.76%**
Macro Precision	0.5367	**0.6333**	**+0.0966**
Macro Recall	0.6500	**0.7000**	**+0.0500**
Macro F1-Score	0.5285	**0.6618**	**+0.1333**
Weighted F1-Score	0.6478	**0.7850**	**+0.1372**

Table 2: Multi-Label Audience Classification Evaluation (Test N = 17):

Metric	Standard LR (Default τ = 0.50)	Class-Weighted LR (Tuned τ = 0.20)	Impact / Trade-off
Exact-Match Accuracy	76.47%	58.82%	-17.65%
Hamming Loss	**0.0980**	0.1765	+0.0785
Macro F1-Score	0.3333	**0.4667**	**+40.0% rel.**
Weighted F1-Score	0.8068	**0.8273**	**+0.0205**

• **Summarization Benchmark (19 Held-Out Notices):** Fine-tuned BART achieved ROUGE-1 of 0.4978 (vs. baseline 0.4539), ROUGE-2 of 0.2954 (vs. baseline 0.2823), and ROUGE-L of 0.4061 (vs. baseline 0.3613). Under Stage 2 token masking, 0 out of 19 generated summaries contained ungrounded or hallucinated URLs.

7. Error Analysis & Key Limitations

1. **Academic Category Small-Sample Vulnerability:** In the training set, academic notices comprise only 8 documents (10%). In the test split (N=17), both academic circulars (`acad-005` regarding Academic Bank of Credits and `acad-019` regarding an industrial visit) resembled general administrative circulars and were classified as announcements, resulting in test recall of 0.00. Adding syllabus updates and curriculum circulars is necessary to bolster minority lexical coverage.
2. **Audience Sparsity & Threshold Trade-off:** In the held-out test split, admin is present in only 2 notices and faculty in 3. Standard logistic regression achieves 76.47% exact-match accuracy trivially by predicting `['student']` for 100% of notices (0% minority recall). Tuning thresholds to τ=0.20 trades exact-match accuracy (58.82%) for a +40% increase in Macro F1 (0.4667), ensuring faculty circulars (e.g., `ann-014`) are successfully flagged.
3. **Pipeline Cascade Risk:** If Stage 1 misclassifies target audiences (e.g., omitting faculty), Stage 3 audience gating triggers an immediate null response (*"No specific faculty action stated."*), bypassing summarization even if actionable text exists.
4. **Deployment Scope:** The asset is currently usable via local Python imports and CLI. An HTTP REST daemon is queued for subsequent milestones.

8. Conclusion

In Week 6, University Communications Intelligence achieved its objective of delivering a functional, reusable NLP asset. By combining cost-sensitive intent classification (82.35% test accuracy) with audience-gated routing and token-masked BART summarization, the system establishes a robust, modular framework for institutional communication automation.